

MCKINNON ST SIMONS ISLAND, GA, USA

WMO: 722137

Lat: 31.152N

Lon: 81.391W

Elev: 16

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13878

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.2	34.7	15.0	11.8	38.6	19.6	14.8	41.5	20.7	50.9	18.8	51.3	6.2	350	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	92.9	77.3	90.6	77.4	88.8	77.3	80.4	87.8	79.7	87.1	78.8	86.0	9.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.4	84.8	77.4	142.8	84.0	76.8	139.8	83.4	43.9	87.9	43.0	86.9	42.2	86.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.2	17.1	15.6	DB	25.8	96.8	3.8	1.6	23.0	97.9	20.8	98.8	18.7	99.7	15.9	100.8
			WB	22.3	82.3	3.5	1.2	19.8	83.1	17.8	83.8	15.8	84.4	13.3	85.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.5	52.0	55.2	60.7	67.5	75.0	80.4	82.4	81.8	78.6	70.8	61.3	55.5	(6)
	DBStd	12.47	8.80	8.27	7.69	6.03	4.72	3.05	2.47	2.77	3.79	6.49	7.67	8.46	(7)
	HDD50	187	85	38	13	0	0	0	0	0	0	0	8	43	(8)
	HDD65	1400	408	284	171	44	2	0	0	0	0	25	158	308	(9)
	CDD50	6942	147	185	345	524	776	911	1005	987	857	644	347	214	(10)
	CDD65	2678	4	11	38	118	313	461	540	522	407	204	46	14	(11)
	CDH74	23389	8	34	133	469	2094	4341	5819	5525	3614	1199	138	15	(12)
	CDH80	7225	0	1	19	73	478	1391	2180	1931	960	178	14	0	(13)
(14) Wind	WSAvg	7.8	7.4	7.7	8.3	8.5	8.8	8.0	7.8	7.5	8.2	7.6	7.4	7.1	(14)
(15) Precipitation	PrecAvg	48.6	3.4	3.7	4.0	2.6	3.1	5.3	5.3	6.3	6.5	3.3	2.5	2.9	(15)
	PrecMax	69.1	9.4	8.5	10.7	9.3	10.3	15.7	12.1	18.1	21.2	11.0	7.4	8.5	(16)
	PrecMin	34.1	0.4	0.7	0.5	0.2	0.3	0.7	1.0	1.2	0.1	0.0	0.2	0.0	(17)
	PrecStd	8.3	2.4	1.8	2.6	1.9	2.3	3.5	2.9	3.3	4.4	2.8	1.9	2.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.9	82.8	87.0	92.3	95.1	95.5	95.4	91.0	86.5	81.4	76.1	(19)
		MCWB	65.9	67.2	67.5	69.9	73.5	76.8	77.8	78.2	76.0	75.3	70.3	68.4	(20)
	2%	DB	70.2	73.5	77.6	81.6	87.8	91.6	92.7	91.8	88.2	84.1	77.5	72.7	(21)
		MCWB	62.9	64.7	65.4	69.0	72.4	76.4	77.9	78.6	76.2	74.1	69.1	66.7	(22)
	5%	DB	67.2	69.9	74.3	78.6	84.6	88.8	90.3	89.4	86.5	81.9	75.0	69.9	(23)
		MCWB	61.7	62.4	64.7	68.6	72.5	76.7	78.1	78.7	76.0	73.6	68.5	65.7	(24)
	10%	DB	64.4	66.6	71.4	76.4	82.3	86.5	88.4	87.7	85.0	80.0	72.7	67.4	(25)
		MCWB	60.2	61.1	63.5	67.7	72.3	76.3	77.9	78.1	75.6	72.5	67.2	64.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.8	69.1	70.5	74.5	77.8	80.7	81.5	82.0	79.7	79.1	74.5	70.8	(27)
		MCDB	72.4	75.5	76.1	79.3	84.4	87.4	89.1	89.6	86.1	82.6	77.4	73.2	(28)
	2%	WB	65.1	66.5	68.4	72.7	76.2	79.4	80.4	80.4	78.4	76.9	72.3	68.5	(29)
		MCDB	68.3	71.2	73.9	77.5	82.4	86.6	88.2	87.9	84.6	81.5	75.1	70.9	(30)
	5%	WB	63.3	64.5	67.0	71.2	75.1	78.3	79.6	79.7	77.8	75.5	70.3	66.5	(31)
		MCDB	66.0	68.1	71.8	75.9	81.1	85.4	87.6	87.1	83.8	80.0	73.4	69.2	(32)
	10%	WB	61.2	62.5	65.4	69.7	74.1	77.4	78.6	78.8	77.1	74.2	68.2	64.3	(33)
		MCDB	64.0	65.7	69.9	74.5	80.0	84.4	86.2	85.8	82.8	78.6	71.8	67.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	16.1	16.0	14.9	13.8	13.1	13.1	12.1	11.5	14.2	16.3	15.7	(35)
		MCDBR	17.5	18.2	18.4	16.3	17.1	16.3	15.5	14.3	13.0	13.3	16.2	15.9	(36)
	5% WB	MCWBR	12.0	11.3	9.5	7.8	7.4	6.9	6.4	5.8	5.5	6.6	9.6	11.0	(37)
		MCWBR	15.3	15.7	15.4	13.5	13.5	14.1	14.0	12.9	11.3	10.8	12.8	14.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.337	0.346	0.377	0.400	0.443	0.491	0.513	0.500	0.429	0.399	0.360	0.350	(40)
		taud	2.530	2.508	2.438	2.365	2.280	2.200	2.147	2.189	2.386	2.433	2.496	2.518	(41)
	Ebn at Noon		283	291	287	283	270	256	250	252	267	268	272	272	(42)
		Edh at Noon	27	30	35	39	43	46	49	46	36	33	28	27	(43)
(44) All-Sky Solar Radiation	RadAvg		964	1158	1531	1897	2049	1964	1963	1786	1511	1328	1062	855	(44)
	RadStd		61	128	144	121	116	145	116	115	128	125	64	69	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		+1.30	+2.57	N/A	+0.72	N/A	N/A	-59	-238	+445	+293	
(47)	Regional (29 neighbors)	+0.88	+1.37	N/A	N/A	N/A	N/A	N/A	-131	+301	+216	(47)

Nomenclature: See separate page